

Fate Hardin

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TECHNICAL SKILLS

Game Engines: Unreal Engine 5/4 (C++ & Blueprints), Unity (C#)

XR & Spatial: Meta Quest, HTC Vive XR Elite, Magic Leap 2, CloudXR, Nvidia Omniverse (Omni Kit), iOS

Multiplayer & Cloud: AWS GameLift, Unity Photon, AWS EC2 & Cognito, Docker

Languages: C++, C#, Blueprints, Python, JavaScript, Java, Swift

AI & Simulation: LangChain, TensorFlow, MediaPipe, OpenCV, LM Studio, RAGFlow

WORK EXPERIENCE

XR Technologist | *Brightline Interactive*

January 2024 – January 2025

- Architected a Magic Leap 2 World Building Plugin for Nvidia Omniverse using Omni Kit — enabling real-time AR spatial primitive manipulation via remote rendering, with matrix-based transforms for accurate 3D placement and scale in the user's physical environment.
- Delivered CloudXR streaming clients for a \$4M+ Department of Defense Spatial Computing platform, streaming cloud-rendered Unreal Engine 5 environments to HTC Vive XR Elite headsets with sub-frame latency over standard network connections.
- Integrated an LLM with RAGFlow as a microservice inside the Omniverse editor, providing context-aware in-editor assistance for environment construction — querying project documentation and asset metadata to surface relevant guidance during authoring.
- Engineered EC2 auto scaling groups with custom AMIs and launch templates to dynamically provision GPU-backed virtual machines, enabling reliable concurrent XR session scaling without manual intervention.

Game Engineer | *Immersive Health Group*

February 2020 – December 2023

- Designed and owned C++ and Blueprint gameplay systems in UE5 for multi-user networked XR simulations — implementing replicated actor components, player state management, and authority-controlled physics interactions across connected clients.
- Built a multiplayer VR medical training simulation with server-authoritative session management via AWS GameLift and Unity Photon, supporting simultaneous deployments across Windows, Mac, and Meta Quest with dynamic matchmaking.
- Engineered a VR clinical procedure training environment in UE4 with a custom 3D inventory system, interactive equipment actors, and procedure-step validation — including a real-time instructor assessment layer for remote performance review and compliance tracking.
- Owned full UI/UX development across UE4/UE5 and Unity, building diegetic in-world and HUD interfaces for XR and desktop; managed complete build, certification, and release pipelines for Meta Quest Store and iOS App Store.

PROJECTS

Gravity & N-Body Simulation | Personal

2024

- Browser-based sandbox visualizing N-body Newtonian gravitational dynamics in Three.js — $O(n^2)$ integrator with an experimental octree partition for Barnes-Hut spatial decomposition and future acceleration of force calculations at scale.

Hand-to-Mouse Gesture Control | Personal

2023

- Trained a CNN in Keras on MediaPipe hand landmark data to classify hand gestures and map them to real-time mouse inputs (left/right click, cursor movement) — OpenCV for live video capture, PyAutoGUI for OS-level mouse control.

Personal Ubuntu Home Server | Personal

2023 – Present

- Self-hosted Linux server running Jellyfin, Samba, and automated iCloud photo backup via icloudpd — secured with WireGuard VPN for encrypted remote access from any network, demonstrating end-to-end infrastructure ownership beyond game development.

EDUCATION

M.S. in Computer Science — Middle Tennessee State University

Expected May 2028

B.S. in Computer Science — University of Tennessee at Martin

May 2020